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ID:

Xi'an Jiaotong-Liverpool University
西交利物浦大学

Module code: CHE105

Module title: Introductory Organic Chemistry for Non-Chemists

Semester 1, 2024/25, Year 2

Coursework 1

Module Examiner: Dr. Eric Amigues

Department of Chemistry, Tel. 0512-88167123

INSTRUCTIONS to CANDIDATES

1. The total number of marks is 100. The numbers in brackets indicate the marks assigned to the question.

2. Answer **ALL** questions

3. Submit your answers as pictures in ONE pdf file and uploaded to Learning Mall Online with the name CHE105_IDNo_Name.

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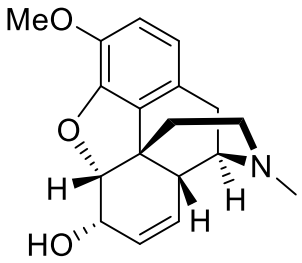
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Problem 1 **[20 marks]**

Below is the structure of the analgesic drug codeine. Please identify and indicate the functional groups present in the molecule (point them out on the structure)

Indicate the hybridization state of all the carbons and their respective geometry.



codeine

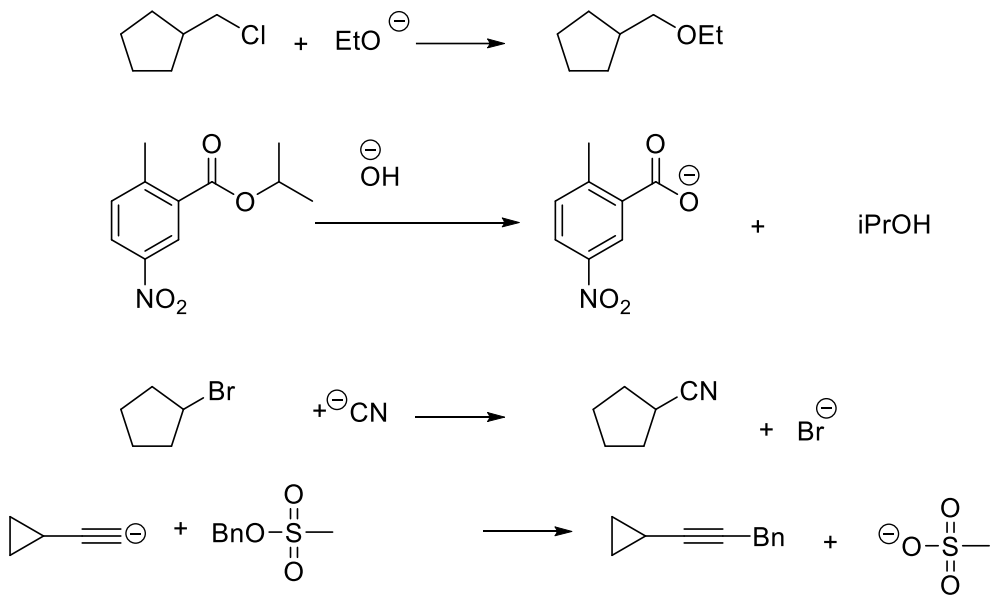
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Problem 2 **[20 marks]**

For each of the following reactions, identify the atoms which are acting as nucleophile, electrophile, and draw a curved-arrow diagram showing a one-step mechanism. (REDRAW THE FULL MECHANISM, DO NOT SIMPLY DRAW ARROWS ON THE SCHEME BELOW)



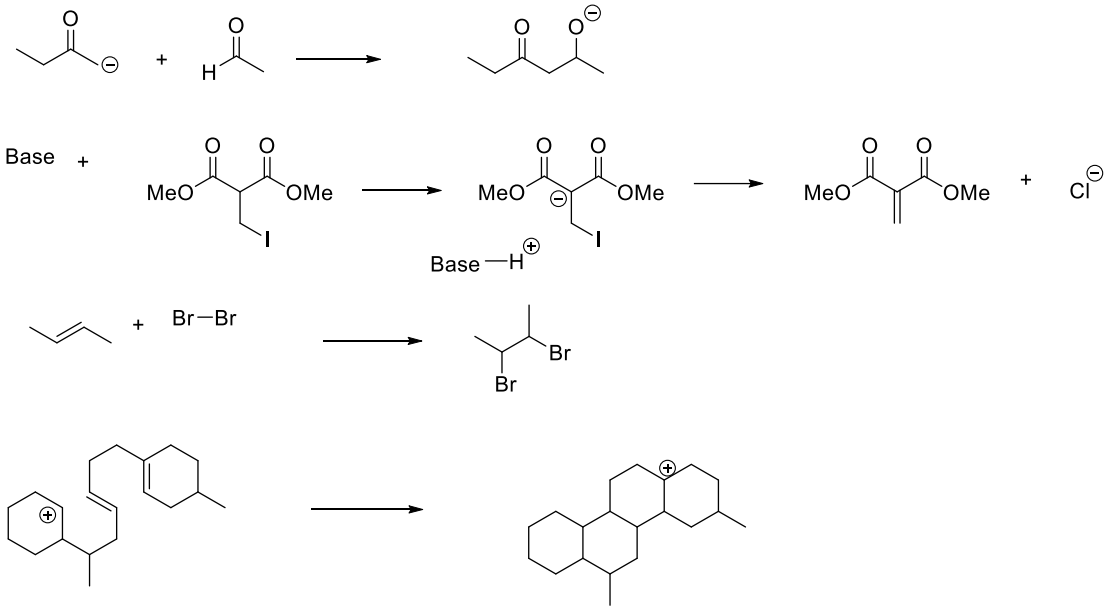
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Problem 3 **[20 marks]**

Complete the following reaction mechanisms, using curly arrows. (REDRAW THE FULL MECHANISM, DO NOT SIMPLY DRAW ARROWS ON THE SCHEME BELOW)



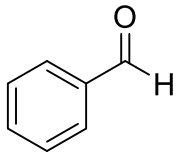
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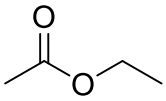
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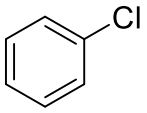
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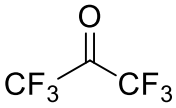
Problem 4 **[15 marks]**

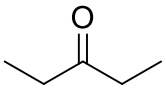
Draw the partial charges of the following molecules.





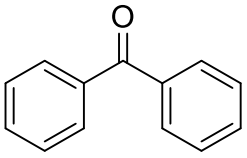


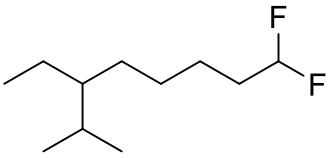


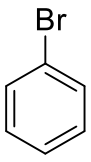


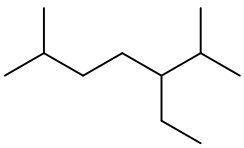
Problem 5 **[15 marks]**

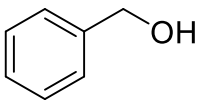
Name the following molecules (using IUPAC nomenclature).











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Problem 6 **[10 marks]**

Draw the structures of the following molecules:

1) 2,4-dinitrotoluene

2) 2,2,2-trifluoro-1-chloro-ethan-1-ol

3) Benzophenone

4) 2-chloro-1-phenylethan-1-ol

5) 3-chloro-2-methyl pentane

6) N,N-dimethylformamide

7) Pyridine

8) piperidine

9) Acetophenone

10) *N,N* dimethylaminopyridine